

- 3 (a)** power is the rate of doing work or $\text{power} = \text{work done} / \text{time (taken)}$ or
power = energy transferred / time (taken) B1 [1]
- (b) (i)** as the speed increases drag / air resistance increases B1
resultant force reduces hence acceleration is less B1
constant speed when resultant force is zero B1 [3]
(allow one mark for speed increases and acceleration decreases)

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- (ii) force from cyclist = drag force / resistive force
 $P = 12 \times 48$
 $P = 576 \text{ W}$ B1
M1
A0 [2]
- (iii) tangent drawn at speed = 8.0 m s^{-1}
gradient values that show acceleration between 0.44 to 0.48 m s^{-2} M1
A1 [2]
- (iv) $F - R = ma$ C1
 $600 / 8 - R = 80 \times 0.5$ [using $P = 576$] $576 / 8 - R = 80 \times 0.5$ C1
 $R = 75 - 40 = 35 \text{ N}$ $R = 72 - 40 = 32 \text{ N}$ A1 [3]
- (v) at 12 m s^{-1} drag is 48 N , at 8 m s^{-1} drag is 35 or 32 N
 R / v calculated as 4 and 4 or 4.4
and consistent response for whether R is proportional to v or not B1 [1]